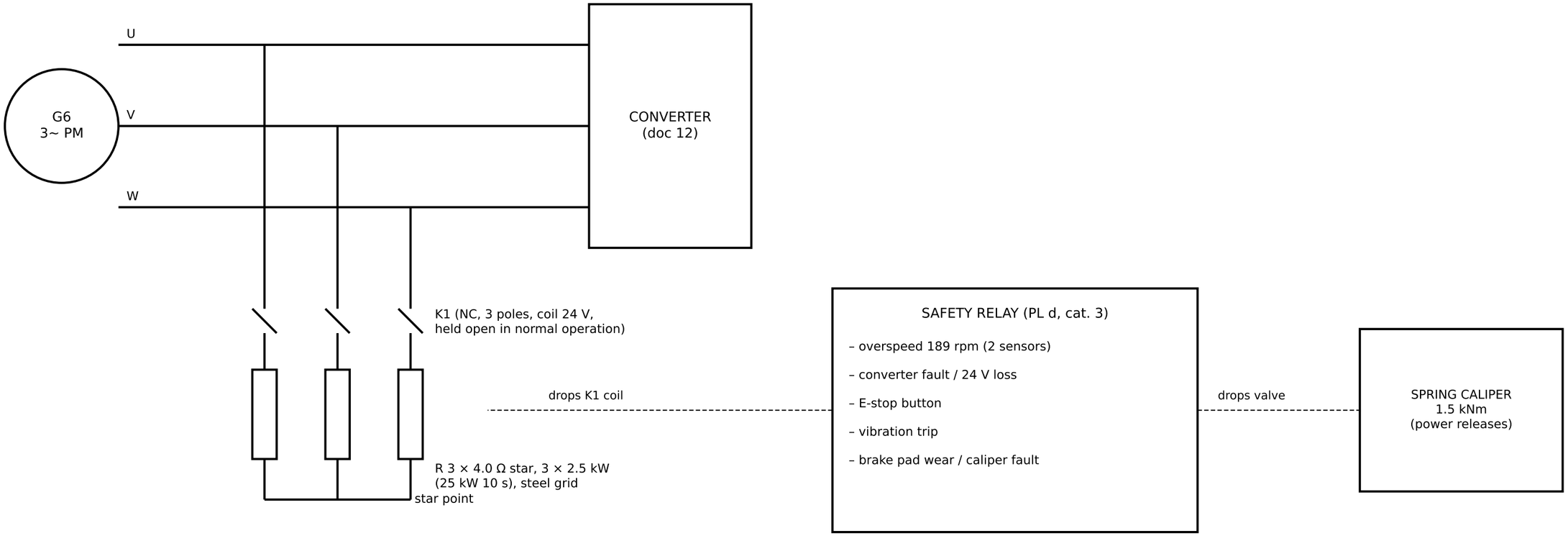
SINGLE-LINE: BOTH STOPPING CHANNELS DROP TOGETHER ON A TRIP



NOTES

1. Doc 29 / 31: K1 is normally closed. While the safety relay is healthy it holds K1 open; any trip or loss of 24 V closes K1 and drops the brake at the same time. Either channel alone keeps the rotor below the trip speed (doc 31: brake alone stops in ≤ 2.4 s, peak 173 rpm; resistor alone slows to a 3–15 rpm crawl).
2. Resistor current 43 A peak at trip speed with hot magnets; energy of one stop ≈ 120 kJ. Grid resistors in a ventilated 1.4301 box (600 × 400 × 250) on the tower base cabinet wall, not in the nacelle.
3. K1: 3-pole contactor with mechanically linked mirror contact (EN 60947-4-1 Annex F) for monitoring; AC-3 rating ≥ 63 A at 690 V.
4. Brake fault alarm: with the brake failed the rotor only crawls, so a failed or worn brake keeps the turbine parked and yawed out until repaired (doc 31 §6).
5. Spring pack stays 1.5 kNm (Martin, 2026-11-06). Parked yaw-lost 50-year peak aero torque 1.45 kNm; if exceeded the rotor creeps slowly against the brake and the closed K1 (doc 32 §1). Use the lock pin NP1-320 for long unattended outages.

Dimensions mm · ISO 2768-mK · ISO 8015 · first angle projection · edges broken 0.5 unless stated · NOT CHECKED: prototype release only after review

6 kW WIND TURBINE S6 (ICELAND) · FIXED PITCH, STALL CONTROL

BRAKING RESISTOR BOX AND SAFETY CHAIN

DRAWING NO.	REV / DATE	SCALE (A3)
NS6-600	A · 2026-11-06	schematic / 1:5
MATERIAL	MASS (CALC.)	DRAWN / CHECKED
Box 1.4301, IP54; resistors steel-grid	≈ 18 kg	Claude / not checked